# Determine the Volume of Rectangular Prisms

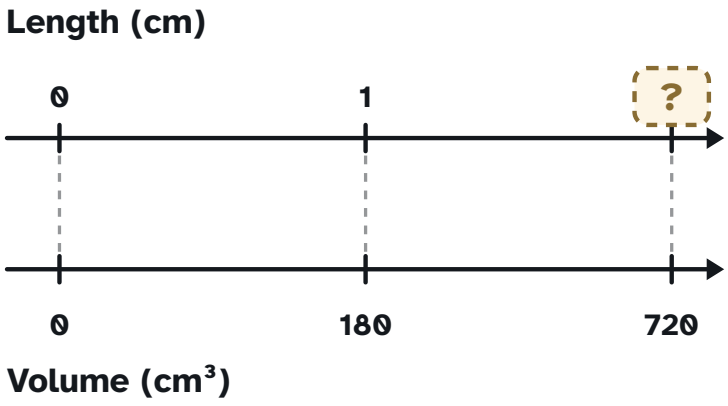
Find a missing edge of a box when you know its volume and two edges.

**Name** _____ **Date** _____

## Tonight's spark

The label on a storage box is torn. It says the box holds 720 cubic centimeters. The box is 15 cm wide and 12 cm tall. How long is the box?

**Guess first: is the box longer or shorter than 15 cm?**




**Our class model: work the formula backwards**

## Try it

Work it out on paper. Write one answer on each line.

1. **A box has a volume of 30, a length of 5 and a width of 3. What do you do first to find the height?**

   **A.** Add 5 and 3 **B.** Divide 30 by 5 only **C.** Multiply 5 × 3 **D.** Subtract 3 from 30

   **My answer** _____

2. **A box has a volume of 24 cm³, a length of 4 cm and a width of 3 cm. What is its height?**

   **A.** 2 cm **B.** 6 cm **C.** 8 cm **D.** 12 cm

   **My answer** _____

**3** **A box has a volume of 72 in³ and a base area of 18 in². What is its height?**

**A.** 54 in. **B.** 4 in. **C.** 90 in. **D.** 1,296 in.

**My answer** _____

**4** **A box is 2 ft long, 3 ft wide and 1.5 ft tall. What is its volume?**

**A.** 6 cubic ft **B.** 12 cubic ft **C.** 6.5 cubic ft **D.** 9 cubic ft

**My answer** _____

**5** **A box has a volume of 45 cm³, a length of 5 cm and a height of 3 cm. What is its width?**

**A.** 9 cm **B.** 3 cm **C.** 15 cm **D.** 30 cm

**My answer** _____

**6** **A box is 4 in. by 2 in. by 2.5 in. If 2.5 is rounded down to 2 before multiplying, how many cubic inches are lost?**

**A.** 4 cubic in. **B.** 0.5 cubic in. **C.** 2 cubic in. **D.** 8 cubic in.

**My answer** _____

### Talk about it

How do you find a missing edge when you know the volume and two other edges?

### Home mission

Pick a small box. Pretend its volume is 100 cubic centimeters and two edges are 5 cm and 4 cm. Can your student find the third edge, then check by multiplying all three?

### If we get stuck, ask

- How many edges does the formula use? Did you use all of them?
- What do the two edges you know multiply to?
- Did you round a fraction? Put the exact value back in.

**Do it on a screen**

Scan or type the link. Same lesson, with hints.

**eduwonderlab.com/lessons/5-5-part2/homework.html**

**How did it feel? (optional)**

Tricky Getting there Got it